



iPhone 13 Battery Replacement

iPhone batteries are rated to hold 80% of their...

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INTRODUCTION

iPhone batteries are rated to hold 80% of their capacity for up to 500 charge cycles, which lasts roughly 18-24 months for most users. After that, your iPhone may need to be charged far more frequently, and iOS may warn you that performance is affected (in other words, your phone will run slower). Use this guide to replace your battery and restore your iPhone to like-new performance.

If your battery is swollen, [take appropriate precautions](#).

Note: In this guide, the loudspeaker is removed to lower the chance of a battery adhesive strip breaking. It's possible to remove the iPhone battery without removing the loudspeaker.

Note: On iOS 17.6 and earlier, your iPhone may display a warning about the “genuineness” of the battery after the repair, even when using original Apple parts. If your iPhone functions normally, you can safely ignore the warning. On iOS 18 and newer, authenticate your new battery using Apple's [Repair Assistant](#).

For optimal performance, after completing this guide, [calibrate](#) your newly installed battery: Charge it to 100% and keep charging it for at least two more hours. Then use your iPhone until it shuts off due to low battery. Finally, charge it uninterrupted to 100%.



TOOLS:

[Clampy - Anti-Clamp](#) (1)
[Hair Dryer](#) (1)
[Heat Gun](#) (1)
[P2 Pentalobe Screwdriver iPhone](#) (1)
[Suction Handle](#) (1)
[iFixit Opening Picks \(Set of 6\)](#) (1)
[Phillips #00 Screwdriver](#) (1)
[Tri-point Y000 Screwdriver](#) (1)
[ESD Safe Blunt Nose Tweezers](#) (1)
[Tweezers](#) (1)
[Spudger](#) (1)
[Isopropyl Alcohol \(90% or Greater\)](#) (1)
[Safety Glasses](#) (1)
[Packing Tape](#) (1)
[Coffee Filters or a lint-free cloth](#) (1)



PARTS:

[iPhone 13 Battery](#) (1)
[iPhone 13 Battery Adhesive Strips](#) (1)

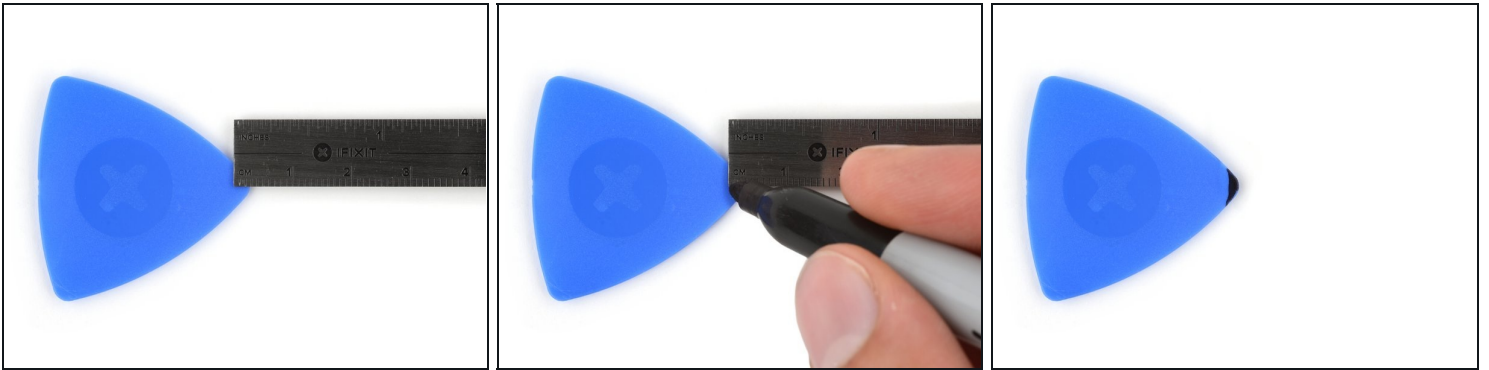
Step 1 — Remove the pentalobe screws



⚠ Caution: Before you begin, discharge your iPhone battery below 25%. A charged lithium-ion battery can catch fire and/or explode if accidentally punctured.

- Power off your iPhone before beginning disassembly.
 - Remove the two 6.8 mm-long pentalobe P2 screws at the bottom edge of the iPhone.
- i** Opening the iPhone's display will compromise its waterproof seals. Have replacement seals ready before you proceed past this step, or take care to avoid liquid exposure if you reassemble your iPhone without replacing the seals.

Step 2 — Mark your opening picks



- ① If inserted too far, an opening pick can damage your device. Follow this step to mark your pick and prevent damage.
- Measure 3 mm from the tip and mark the opening pick with a permanent marker.
 - ① You can also mark the other corners of the pick with different measurements.
 - ① Alternatively, [tape a coin to a pick](#) 3 mm from the tip.

Step 3 — Tape over any cracks



- ① If your iPhone has a cracked screen, keep further breakage contained and prevent bodily harm during your repair by taping over the glass.
- Lay overlapping strips of clear packing tape over the iPhone's screen until the whole face is covered.
- ⚠ Wear safety glasses to protect your eyes from any glass shaken free during the repair.
- If you can't get the suction cup to stick in the next few steps, fold a strong piece of tape (such as duct tape) into a handle and lift the screen with that instead.
- ① If all else fails, you can superglue the suction cup to the screen.

Step 4 — Anti-Clamp instructions



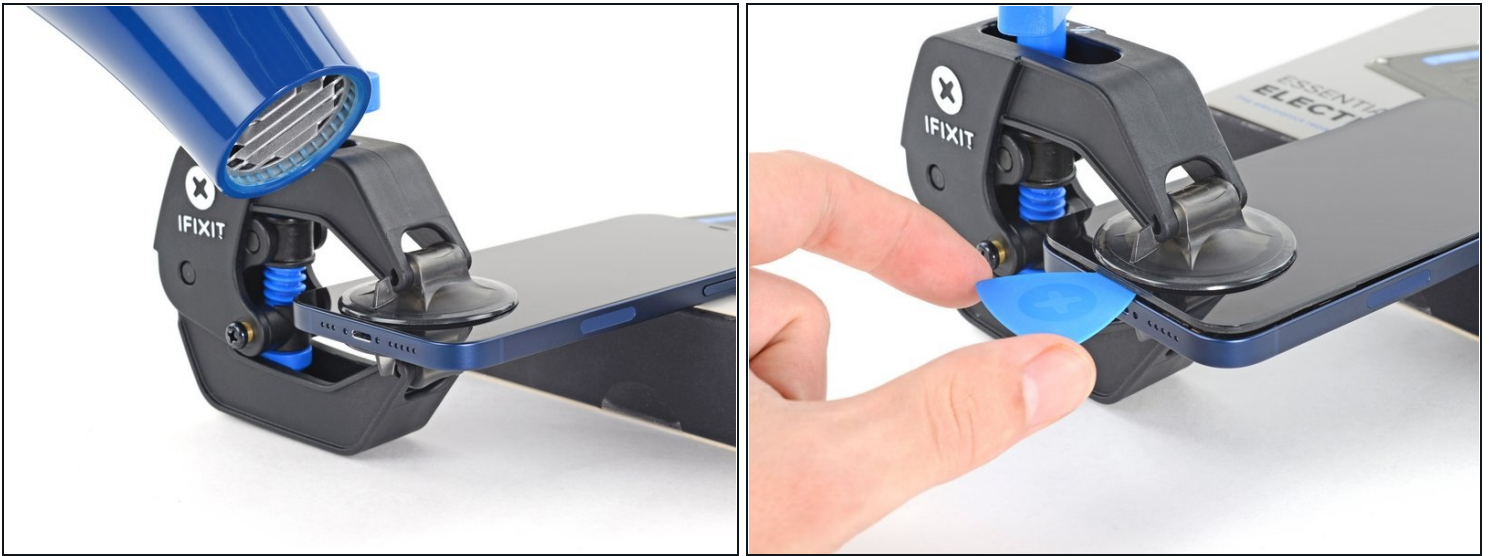
- ① The next three steps demonstrate [the Anti-Clamp](#), a tool we designed to make the opening procedure easier. **If you aren't using the Anti-Clamp, skip down three steps for an alternate method.**
- ① For complete instructions on how to use the Anti-Clamp, [check out this guide](#).
- Pull the blue handle backwards to unlock the Anti-Clamp's arms.
- Slide the arms over either the left or right edge of your iPhone.
- Position the suction cups near the bottom edge of the iPhone—one on the front, and one on the back.
- Squeeze the cups together to apply suction to the desired area.
- ① If you find that the surface of your iPhone is too slippery for the Anti-Clamp to hold onto, you can [use tape](#) to create a grippier surface.

Step 5



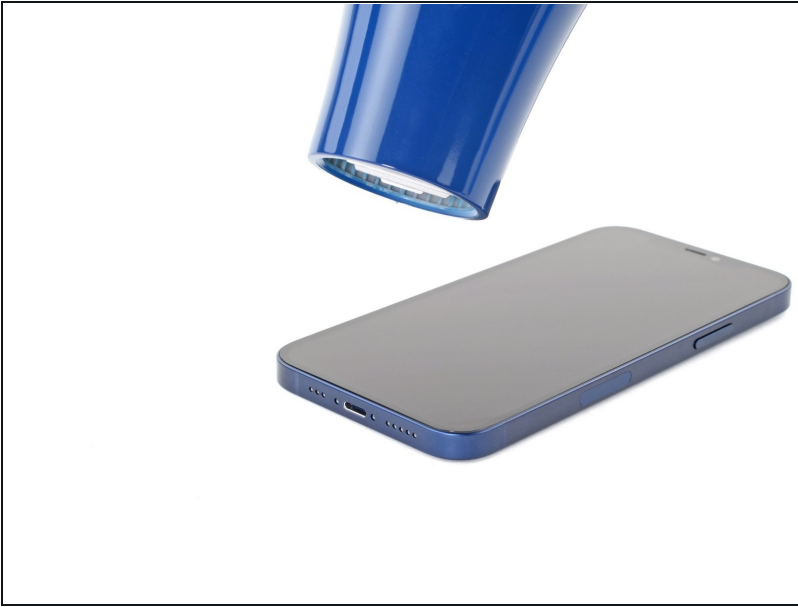
- Pull the blue handle forwards to lock the arms.
- Turn the handle clockwise 360 degrees or until the cups start to stretch.
- Make sure the suction cups remain aligned with each other. If they begin to slip out of alignment, loosen the suction cups slightly and realign the arms.

Step 6



- Support your iPhone on something sturdy such as a hardcover book so it rests hands-free and parallel to the ground; this makes it easier to work with in the following steps.
- Use a hair dryer to heat along the bottom edge of the iPhone. The screen should feel slightly too hot to touch.
 - ① For complete instructions on how to use a hair dryer, [check out this guide](#).
- Wait one minute to give the adhesive a chance to release and present an opening gap.
- Insert an opening pick under the screen's plastic bezel when the Anti-Clamp creates a large enough gap.
 - ① If the Anti-Clamp doesn't create a sufficient gap, apply more heat to the area and rotate the handle a quarter turn.
- ⚠ **Don't crank more than a quarter turn at a time, and wait one minute between turns. Let the Anti-Clamp and time do the work for you.**
- **Skip the next three steps.**

Step 7 — Heat the bottom edge



- Use a [hair dryer](#) or a [heat gun](#) to heat the bottom edge of the iPhone for 90 seconds or until the screen is slightly too hot to touch.

⚠ Never leave your hair dryer in one spot for an extended period of time.

Step 8 — Apply a suction cup



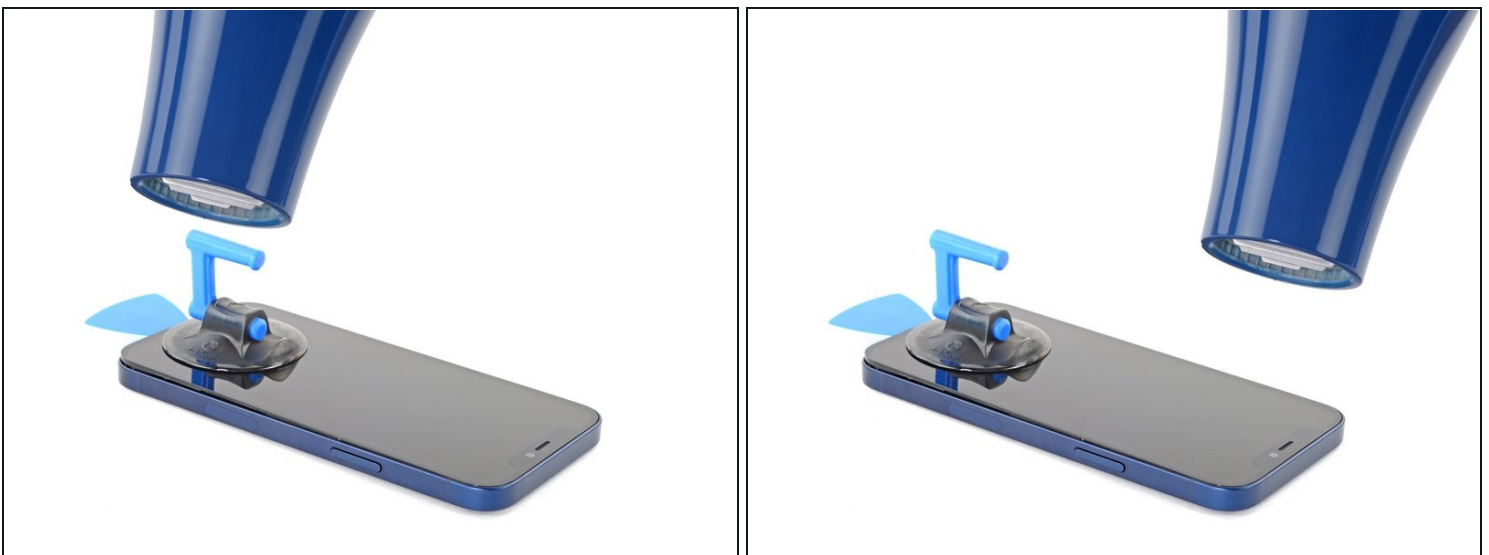
- If you're using a single suction handle, apply it to the bottom edge of the phone, while avoiding the very edge of the glass.

Step 9 — Lift the screen slightly



- Pull up on the suction cup with firm, constant pressure to create a slight gap between the screen and the frame.
 - Insert an opening pick into the gap [under the screen's plastic bezel](#).
- ⓘ The watertight adhesive holding the screen in place is very strong; creating this initial gap takes a significant amount of force. If you're having a hard time opening a gap, apply more heat, and gently rock the screen up and down to weaken the adhesive until you create enough of a gap to insert your tool.

Step 10 — Heat the right edge



- Use a hair dryer to heat the right edge of the iPhone (the edge with the power button) for 90 seconds or until the screen is slightly too hot to touch.

Step 11 — Separate the right adhesive



- Slide the opening pick around the bottom right corner of the iPhone to separate its adhesive.
- Continue sliding up the right edge of the iPhone until you reach the top right corner.
⚠ Don't insert your pick more than 3 mm, as you may damage internal components.
- Leave the pick in the top right corner before continuing.

Step 12



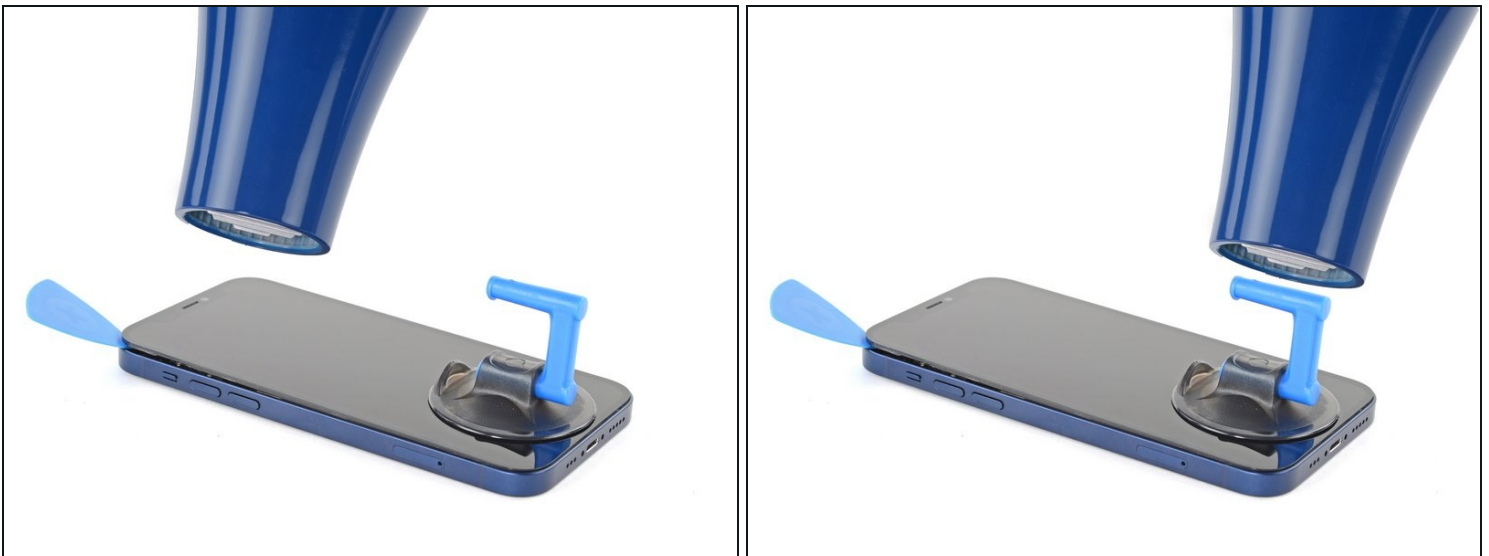
- Use a hair dryer to heat the top edge of the iPhone for 90 seconds or until the screen is slightly too hot to touch.

Step 13



- Slide the opening pick around the top right corner of the iPhone to separate the top adhesive.
⚠ Don't insert your pick more than 3 mm, as you may damage the top sensors.
- Leave the pick in the top left corner before continuing.

Step 14 — Heat the left edge



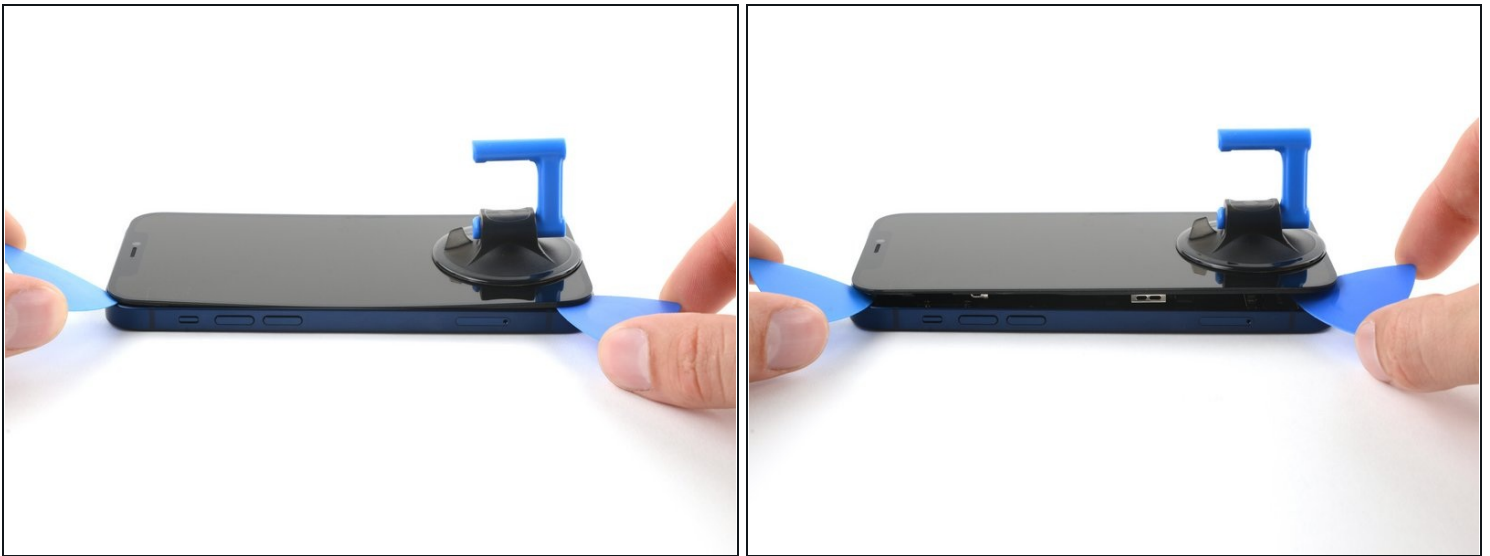
- Use a hair dryer to heat the left edge of the iPhone for 90 seconds or until the screen is slightly too hot to touch.

Step 15 — Screen information



⚠ There are delicate cables along the left edge of your iPhone. **Don't insert your pick here,** as you may damage the cables.

Step 16 — Separate the left adhesive



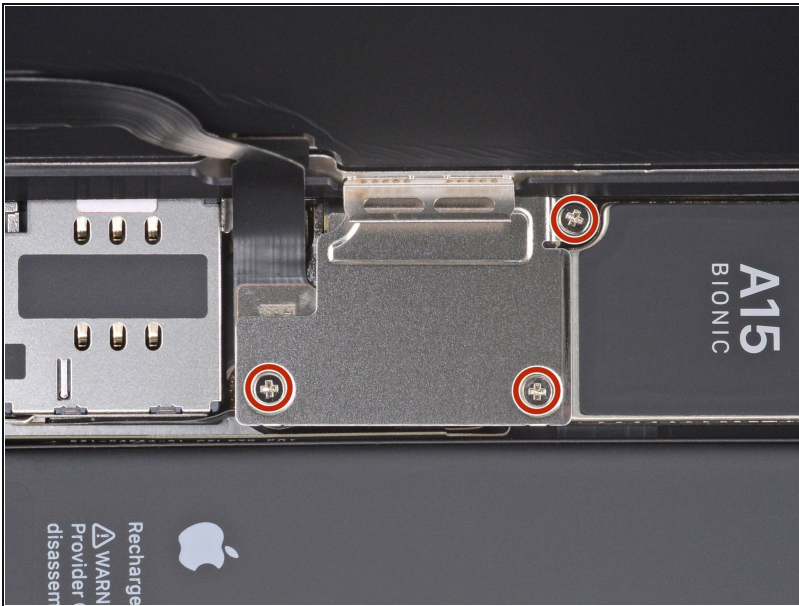
- Insert a second opening pick in the bottom left corner of the iPhone.
- Twist both picks simultaneously until the left edge clips release.
 - ⓘ Apply the twisting force gradually.
 - ⓘ If you're having trouble separating the adhesive, reheat the left edge.

Step 17 — Open the iPhone



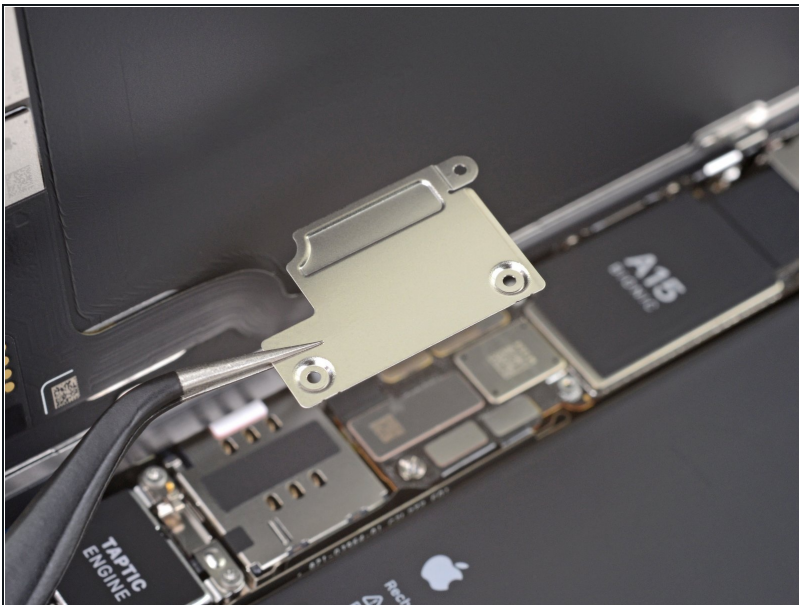
- If you haven't removed your suction handle, remove it now.
 - Rotate your iPhone so the right edge faces you.
 - Open the iPhone by swinging the display up from the right side, like the front cover of a book.
- ⚠ Don't try to fully separate the display yet, as several fragile ribbon cables still connect it to the iPhone's logic board.**
- Lean the display against something to keep it propped up while you're working on the phone.
- ✦** During reassembly, lay the display in position, align the clips along the top edge, and carefully press the top edge into place before snapping the rest of the display down. If it doesn't click easily into place, check the condition of the clips around the perimeter of the display and make sure they aren't bent.

Step 18 — Unscrew the battery and display connector cover



- Use a Phillips screwdriver to remove the three 1.3 mm screws securing the battery and display connector cover.
- ⓘ Throughout this repair, [keep track of each screw](#) and make sure it goes back exactly where it came from to avoid damaging your iPhone.
- ★ During reassembly, this is a good point to power on your iPhone and test all functions before you seal the display in place. Be sure to power your iPhone back down completely before you continue working.

Step 19 — Remove the battery and display connector cover



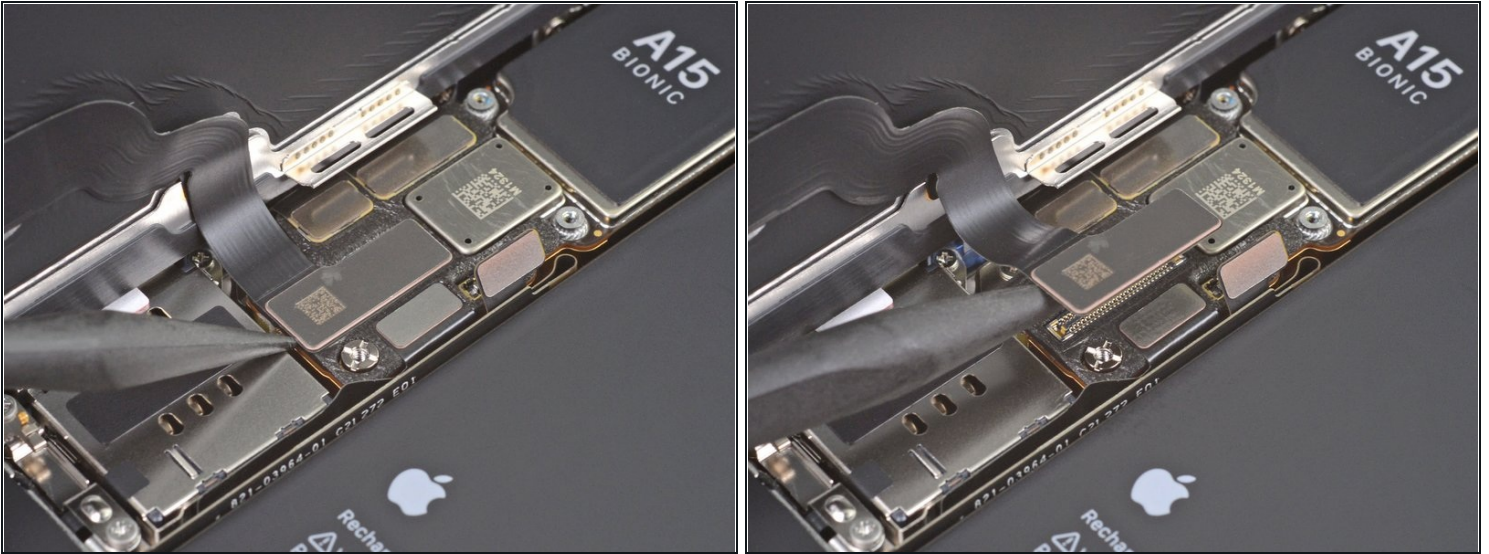
- Remove the cover.

Step 20 — Disconnect the battery



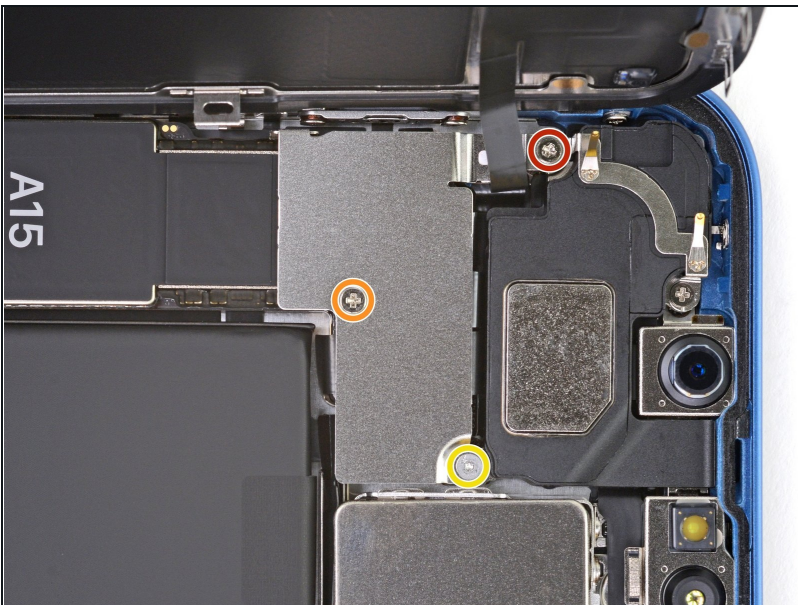
- Use a spudger or a clean fingernail to pry the battery connector up from its socket on the logic board.
 - ① Try not to damage the black silicone seal surrounding this and other board connections. These seals provide extra protection against water and dust intrusion.
- Bend the connector slightly away from the logic board to prevent it from accidentally making contact with the socket and providing power to the phone during your repair.

Step 21 — Disconnect the display cable



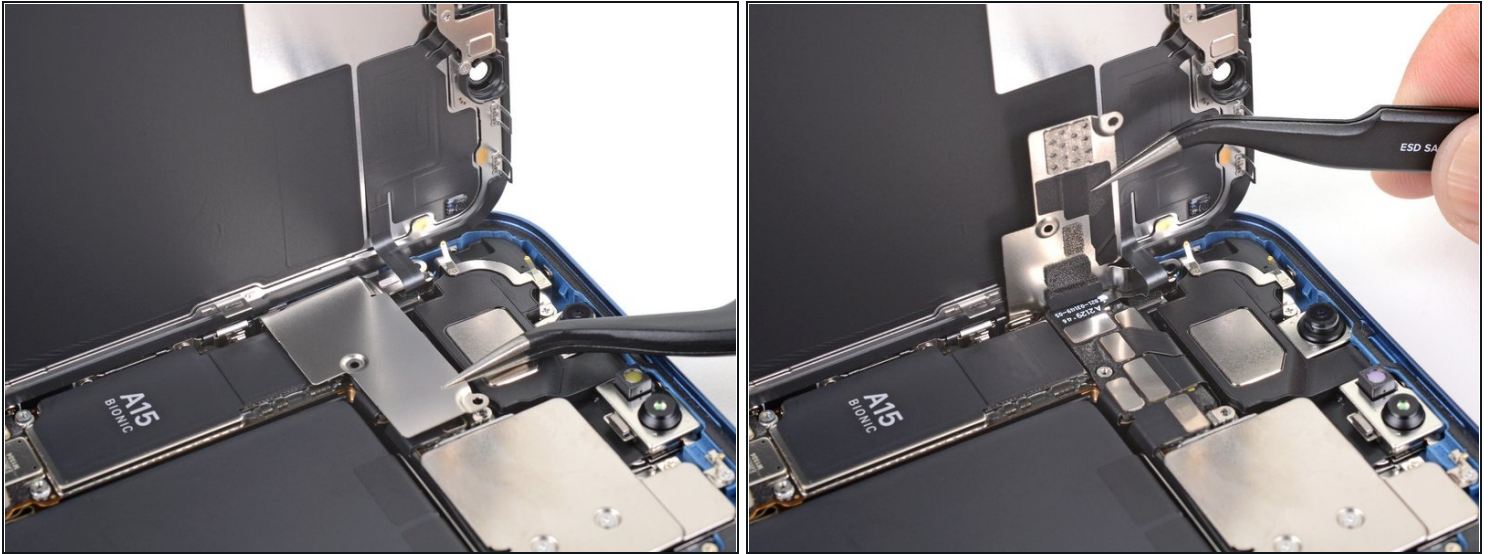
- Use a spudger or a fingernail to disconnect the display cable connector.
 - ① To re-attach [press connectors](#) like this one, carefully align and press down on one side until it clicks into place, then repeat on the other side. Do not press down on the middle. If the connector is misaligned, the pins can bend, causing permanent damage.

Step 22 — Unscrew the front sensor connector cover



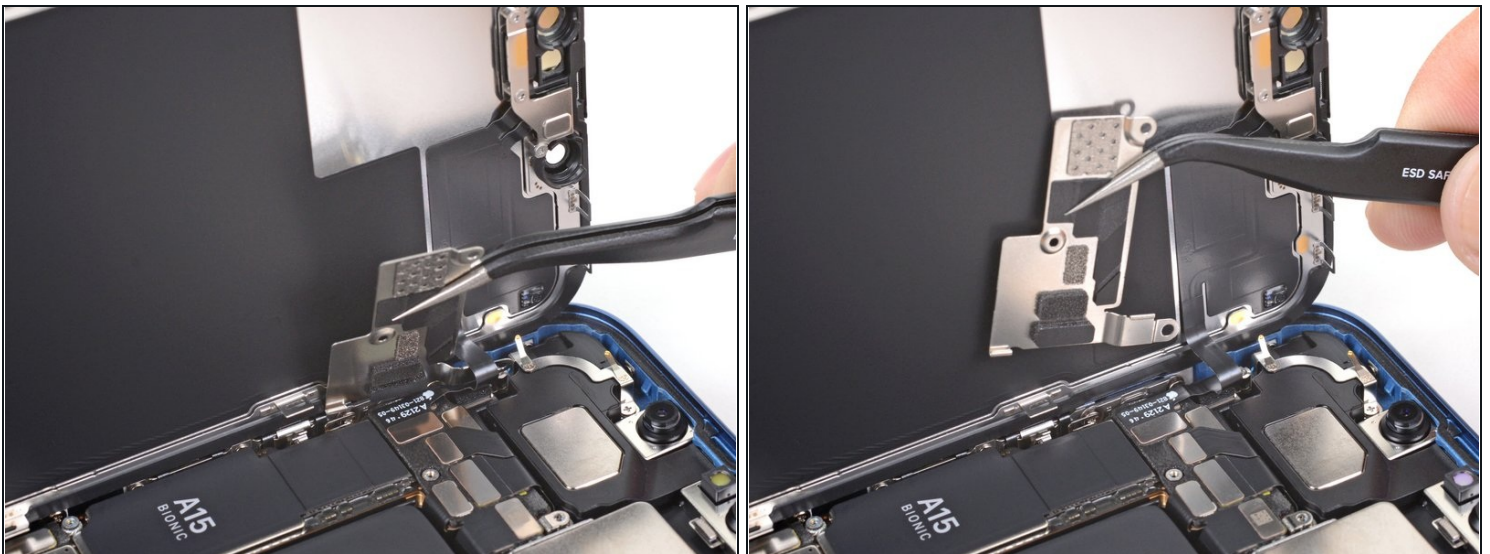
- Remove three screws securing the front sensor connector cover:
 - One 1.7 mm Phillips screw
 - One 1.3 mm Phillips screw
 - One 1.8 mm Y000 screw

Step 23 — Remove the front sensor connector cover



- Swing the cover up from the center of the phone until the small clip by the edge of the phone unhooks.

Step 24



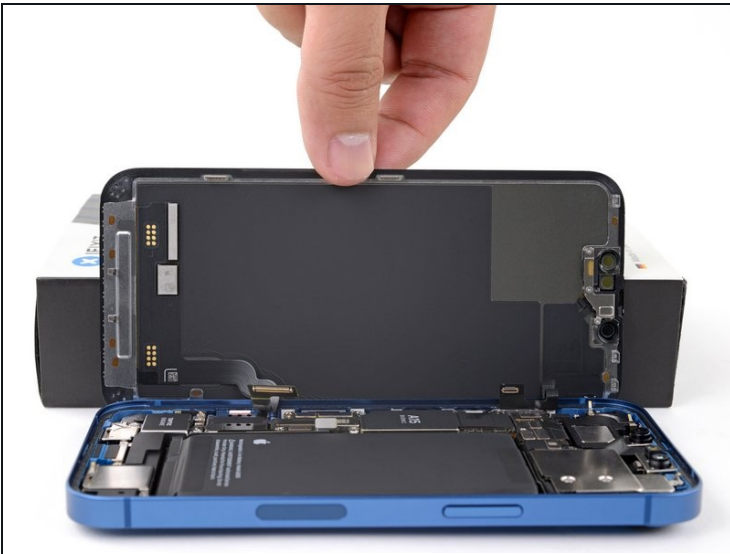
- Remove the cover.
- ☒ During reassembly, insert the [clip on the left edge](#) into its slot first, then hinge the rest of the cover into place.

Step 25 — Disconnect the front sensors



- Use a spudger or a fingernail to disconnect the front sensor assembly cable connector.

Step 26 — Remove the screen



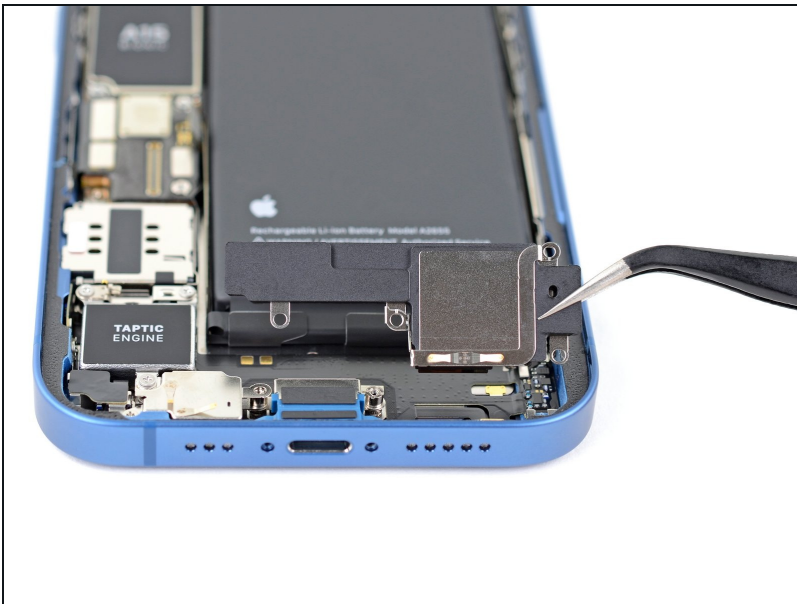
- Remove the screen.
- ☒ During reassembly, pause here if you wish to [replace the waterproof adhesive around the edges of the display](#).

Step 27 — Unscrew the loudspeaker



- Use a Phillips driver to remove the four 1.7 mm screws securing the loudspeaker.

Step 28 — Remove the loudspeaker



- Remove the loudspeaker from the iPhone.

Step 29 — Locate the adhesive pull-tabs



- ① The battery is secured to the rear case with four pieces of stretch-release adhesive—two on the bottom edge of the battery, and two on the top edge.
 - Each piece of adhesive has a black pull-tab at the end, which is lightly adhered to the edge of the battery.
- ① In the following steps, you'll pull each tab to slowly stretch out the adhesive underneath the battery. These "command-strip" style adhesives lose their stick when stretched, so you can remove the battery with ease.
- ① If the strips break, don't panic! They don't always work as intended. Keep reading for tips on removing broken strips.

Step 30 — Remove the stretch-release adhesive



⚠ Caution: Don't jab the battery with any sharp tools. A punctured battery may leak dangerous chemicals or catch fire.

- Use the flat end of a spudger to gently lift up the edge of the top right battery adhesive pull-tab so that you can grab it with a pair of [blunt tweezers](#).
- Peel down the battery adhesive pull-tab to un-stick it from the top edge of the battery.
- Repeat for the top left battery adhesive pull-tab.

Step 31



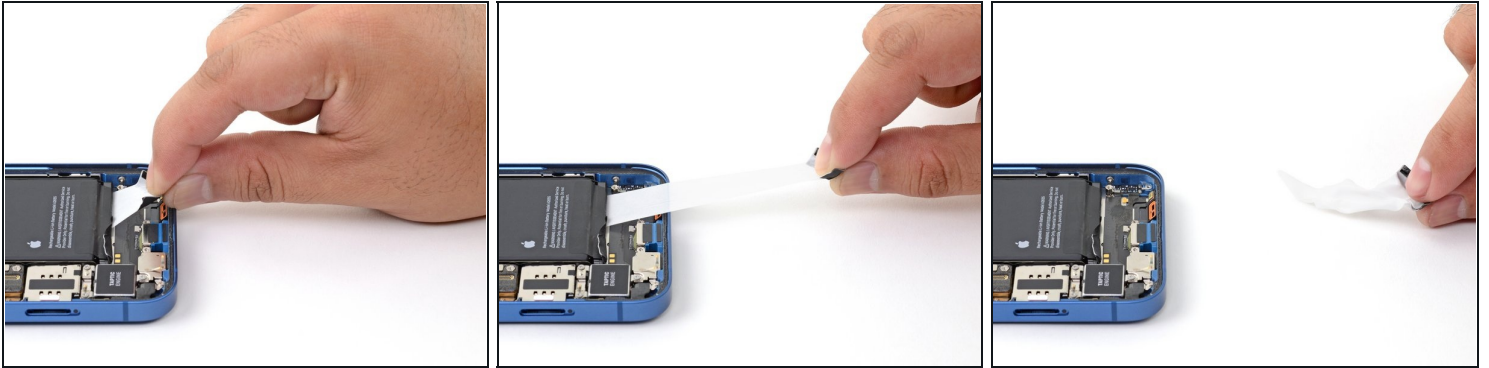
- Peel up the two remaining battery adhesive pull-tabs to un-stick them from the bottom edge of the battery.

Step 32



- ① Optionally, tilt the bottom of the phone up and apply a few drops of high-concentration (90% or higher) isopropyl alcohol to the bottom edge of the battery, and wait a minute. It'll help soften the adhesives, especially if the phone is old.

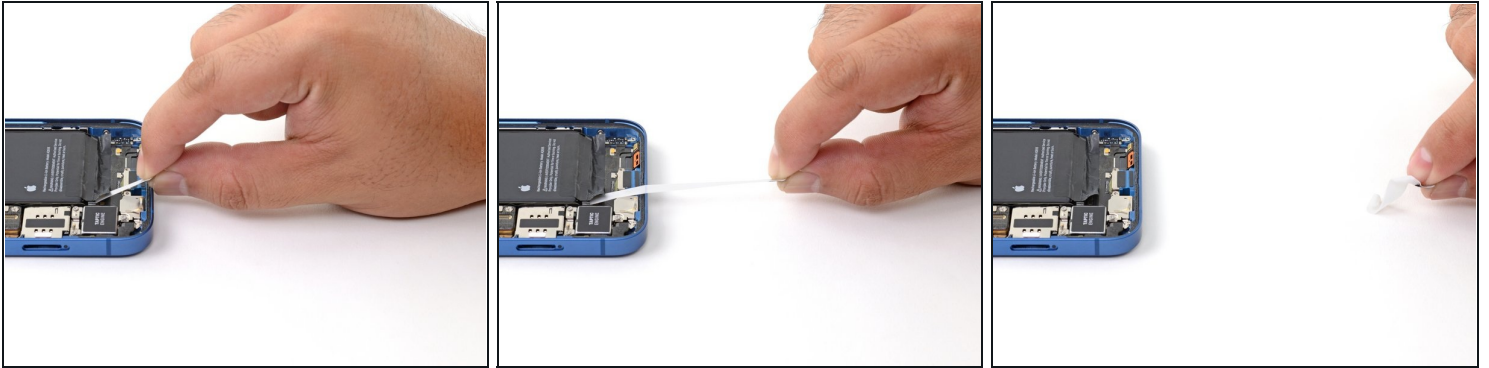
Step 33



- Grab the bottom right pull-tab with your fingers and slowly pull it away from the battery, toward the bottom of the iPhone.
- Pull hard enough to maintain tension on the strip, but don't force it. Give it plenty of time to stretch and un-stick from under the battery.
- Don't press down on the battery. Hold the iPhone firmly by its sides.
- Keep the strip flat and unwrinkled. Try to pull evenly on the whole strip, rather than pulling mainly in the middle or on one side.
- Pull at a low angle so the strip doesn't snag on the edge of the battery.
- If the adhesive strip breaks off, try to retrieve it using your fingers or blunt tweezers, and continue pulling—but do not pry under the battery.

⚠ If any of the adhesive strips break off underneath the battery and can't be retrieved, try to remove the other strips, and then proceed as instructed below.

Step 34



- Grab the bottom left pull-tab with your fingers and pull it away from the battery, toward the bottom of the iPhone.
- Follow the same procedure as with the previous strip: don't press down on the battery, keep the strip flat and unwrinkled, and pull at a low angle.
- If the adhesive strip breaks off, try to retrieve it using your fingers or blunt tweezers, and continue pulling—but **do not** pry under the battery.

⚠ If the adhesive breaks off underneath the battery and can't be retrieved, continue with the next step.

Step 35



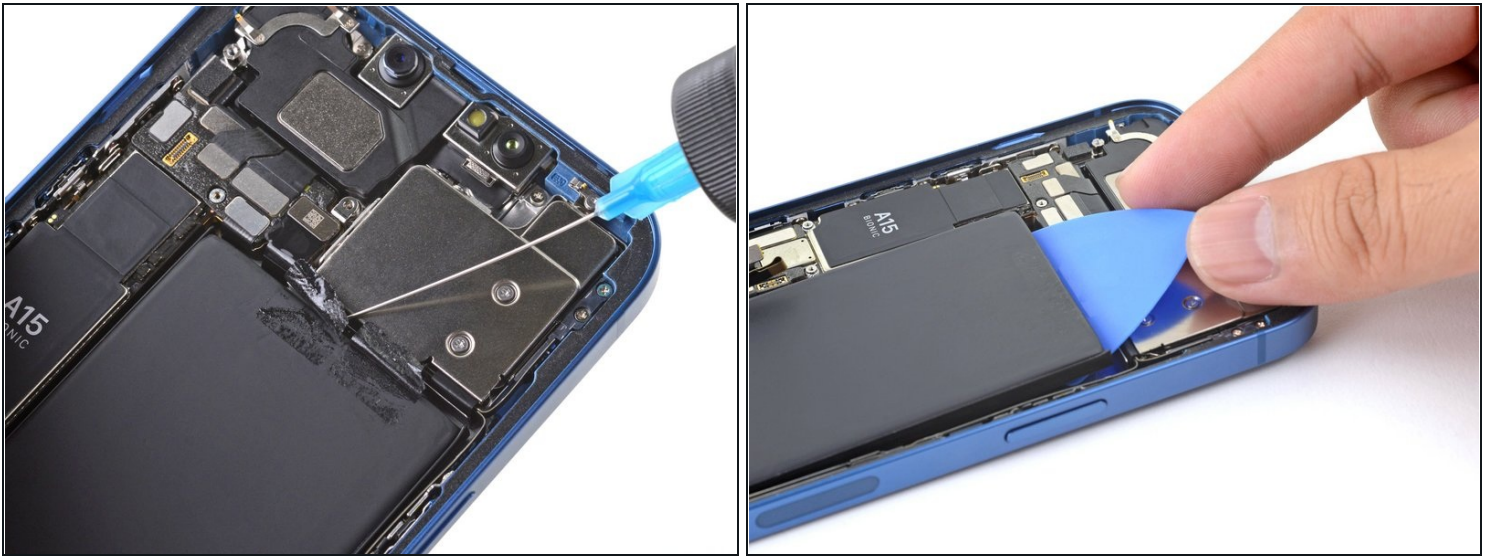
- ① You can optionally wrap the pull-tabs around the tips of blunt tweezers to make it easier to pull out the adhesive strip:
- Use a pair of blunt tweezers to grab the entire width of the pull-tab.
 - Wrap the pull-tab and adhesive strip around the tips of the blunt tweezers.
 - Using the same procedure as before, pull the top right pull-tab to stretch and remove the adhesive strip. Try not to snag the adhesive on anything.

Step 36



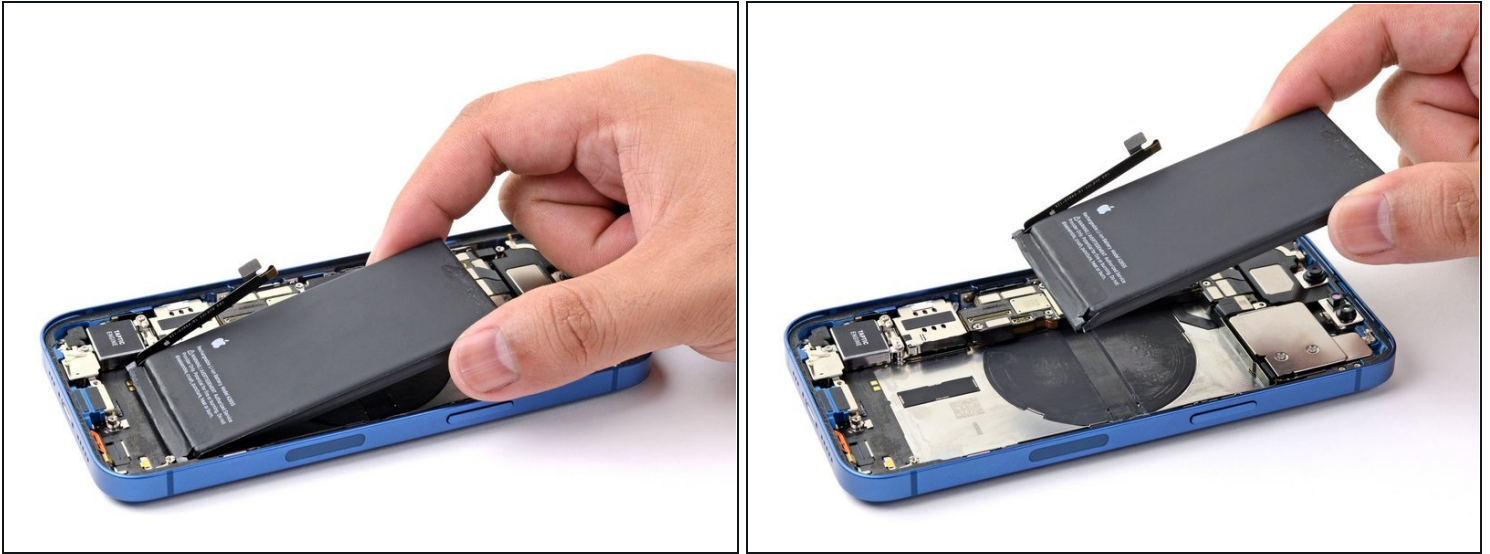
- Lightly place two fingers on the center of the battery to prevent it from flinging out while removing the final adhesive strip.
⚠ Don't press down on the battery with excessive force.
- Use a pair of blunt tweezers to pull and stretch the remaining tab to remove the final adhesive strip holding the battery in place.
 - **If you removed all the adhesive strips successfully, skip the next step.**
 - Otherwise, **continue with the next step below.**

Step 37 — How to remove a stuck battery



- If you're still having trouble removing the battery, apply a few drops of high concentration (90% or higher) isopropyl alcohol under the edge of the battery in the area of the broken adhesive strip(s).
 - Tilt the phone up so the isopropyl alcohol can flow toward the adhesive strip.
 - Wait about one minute for the alcohol solution to weaken the adhesive.
 - Use an opening pick or the flat end of a spudger to gently lift the battery.
- ⚠ Don't try to forcefully lever the battery out. If needed, apply a few more drops of alcohol to further weaken the adhesive. Never deform or puncture the battery with your pry tool.

Step 38 — Remove the battery



- Remove the battery from the iPhone.
- If there's any alcohol solution remaining in the phone, carefully wipe it off with a coffee filter or allow it to air dry before installing your new battery.
- ☑ If your new battery didn't come with adhesive preinstalled, refer to [this guide](#) for help replacing the adhesive strips.
- ☑ Perform a [force restart](#) after reassembly. This can prevent several issues and simplify troubleshooting.

Compare your new replacement part to the original part—you may need to transfer remaining components or remove adhesive backings from the new part before installing.

To reassemble your device, follow these instructions in reverse order.

After your repair, depending on the replacement part, you may be able to [calibrate it using Apple's Repair Assistant](#), available as of iOS 18. Update your device, navigate to Settings → General → About → Parts & Service History, tap Restart & Finish Repair, and follow the onscreen prompts.

Take your e-waste to an [R2 or e-Stewards certified recycler](#).

Repair didn't go as planned? Try some [basic troubleshooting](#), or ask our [iPhone 13 Answers community](#) for help.